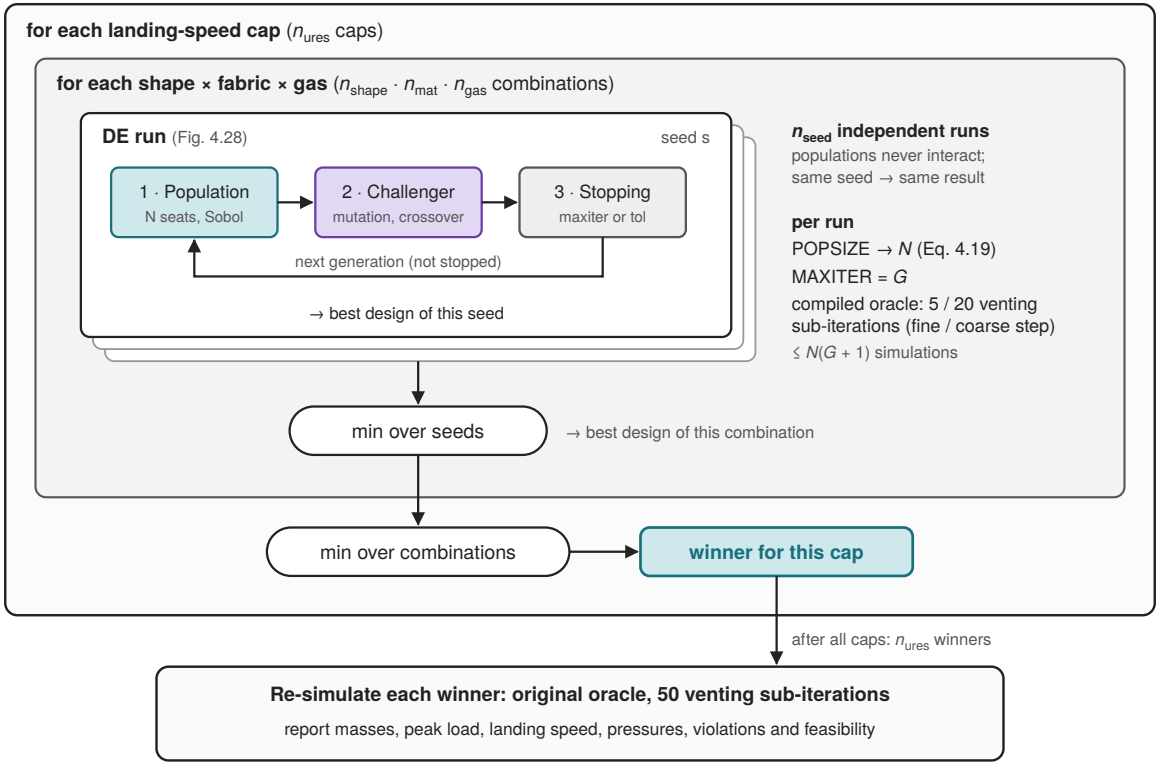
for each landing-speed cap (n_{ures} caps)
for each shape × fabric × gas ($n_{shape} \cdot n_{mat} \cdot n_{gas}$ combinations)
DE run (Fig. 4.28)
seed s
1 · Population
N seats, Sobol
2 · Challenger
mutation, crossover
3 · Stopping
maxiter or tol
next generation (not stopped)
→ best design of this seed
n_{seed} independent runs
populations never interact;
same seed → same result
per run
POPSIZE → N (Eq. 4.19)
MAXITER = G
compiled oracle: 5 / 20 venting
sub-iterations (fine / coarse step)
≤ $N(G + 1)$ simulations
min over seeds
→ best design of this combination
min over combinations
winner for this cap
after all caps: n_{ures} winners
Re-simulate each winner: original oracle, 50 venting sub-iterations
report masses, peak load, landing speed, pressures, violations and feasibility
sweep: $n_{ures} \cdot n_{shape} \cdot n_{mat} \cdot n_{gas} \cdot n_{seed}$ DE runs, up to that × $N(G + 1)$ simulations